

SECTION A

Answer all questions in the spaces provided.

1. Using **outer** electrons only, draw a dot and cross diagram to show the bonding in calcium bromide. [2]

2. Identify the **two** elements from the following list that together produce the most ionic bond. Explain your choice.

bromine

magnesium

oxygen

sodium

[2]

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3. (a) State how a coordinate bond differs from a covalent bond. [1]

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- (b) Give **one** example of a species containing a coordinate bond. [1]

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4. There are 33 known isotopes of krypton. One of the isotopes – ^{81}Kr – decays by electron capture.

(a) Write an equation to show this decay. [1]

- (b) Krypton-81 has been used for dating old groundwater. It takes 6.87×10^5 years for 2.0 g of ^{81}Kr to decay to 0.25 g. Calculate its half-life. [1]

Half-life = years

5. Boron has a relative atomic mass of 10.8. It has only two naturally-occurring isotopes, one of which has an abundance of 80.0% and a mass of 11. Calculate the mass of the second isotope. [2]

Mass =

